Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY – PUNE

CS2311: SOFTWARE ENGINEERING

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ASSIGNMENT 8

PROBLEM STATEMENT

- Perform software cost estimation exercises using Function Point Analysis and Use Case Points for a given software project.

FUNCTION POINT ANALYSIS (FPA)

Step 1: Identify the Five Function Types

1. External Inputs (EI)

```
Weights: Simple=3, Average=4, High=6
```

#	External Input	Complexity	Weight
1	Submit Transaction for Fraud Check	High	6
2	Start/Stop Simulation	Simple	3
3	Update Risk Threshold Settings	Average	4
4	Update Alert Configuration	Average	4
5	Analyst Case Action (Dismiss/Confirm)	Simple	3
6	Device Fingerprint Data	High	6
7	IP Geolocation Data Input	Average	4
8	Merchant Data Registration	Average	4
	Total EI		**34**

2. External Outputs (EO)

```
Weights: Simple=4, Average=5, High=7
```

#	External Output	Complexity	Weight
1	Fraud Decision (BLOCK/CHALLENGE/ALLOW)	High	7
2	XAI Explanation Report	High	7
3	Real-time Alert (Email/SMS/Slack)	High	7
4	Transaction Analytics Report	Average	5
5	Fraud Ring Detection Report	High	7
6	Performance Metrics Dashboard	Average	5
7	Regulatory Compliance Report	Average	5
8	Case Management Export	Average	5
	Total EO		**48**

3. External Inquiries (EQ)

```
Weights: Simple=3, Average=4, High=6
```

#	External Inquiry	Complexity	Weight
1	Get Transaction by ID	Simple	3
2	Get Recent Transactions List	Average	4
3	Get System Statistics	Average	4
4	Get Alerts Feed	Simple	3
5	Get Graph Analytics (fraud rings)	High	6
6	Get Merchant Reputation Data	Average	4
7	Get Performance Metrics	Average	4
8	Get Case Queue for Analyst	Simple	3
9	Get Explainability for Transaction	High	6
	Total EQ		**37**

4. Internal Logical Files (ILF)

```
Weights: Simple=7, Average=10, High=15
```

#	Internal Logical File	Complexity	Weight
1	Transactions Table (SQLite)	High	15
2	User Behavioral Profiles (JSON)	High	15
3	Fraud Alerts Table	Average	10
4	Case Management Records	Average	10
5	Device Reputation Registry	Average	10
6	Merchant Reputation Table	Simple	7
7	Graph Nodes/Edges (fraud network)	High	15
8	ML Model Files (joblib artifacts)	Simple	7
	Total ILF		**89**

5. External Interface Files (EIF) — Weights: Simple=5, Average=7, High=10

#	External Interface File	Complexity	Weight
1	IP Geolocation Database	Average	7
2	Kaggle UPI Dataset (training data)	Average	7
3	PaySim Dataset (training data)	Average	7
4	RBI Regulatory Rules Reference	Simple	5
5	Email/SMS/Slack Webhook APIs	Simple	5
	Total EIF		**31**

Step 2: Unadjusted Function Points (UFP)

Type	Count	Weighted Total
External Inputs (EI)	8	34
External Outputs (EO)	8	48
External Inquiries (EQ)	9	37
Internal Logical Files (ILF)	8	89
External Interface Files (EIF)	5	31
UFP Total	**38**	**239**

Step 3: Value Adjustment Factor (VAF)

Each of the 14 General System Characteristics (GSC) rated **0–5**:

#	General System Characteristic	Rating
1	Data Communications	5
2	Distributed Data Processing	2
3	Performance	5
4	Heavily Used Configuration	3
5	Transaction Rate	5
6	Online Data Entry	4
7	End-User Efficiency	3

8	Online Update	5
9	Complex Processing	5
10	Reusability	3
11	Installation Ease	2
12	Operational Ease	3
13	Multiple Sites	1
14	Facilitate Change	3
TDI		**49**

```
VAF = 0.65 + (0.01 × TDI) = 0.65 + 0.49 = 1.14
```

Step 4: Adjusted Function Points (AFP)

```
AFP = UFP × VAF = 239 × 1.14 = 272.5 ≈ 273 Function Points
```

Step 5: Effort & Cost Estimation

Metric	Value
Adjusted FP	273
Effort at 8 hrs/FP	**2,184 person-hours**
Effort at 10 hrs/FP	**2,730 person-hours**
Team Size	2 developers
Duration	~12 weeks = 480 hrs/person
Cost @ ₹500/hr blended rate	**₹10.9L – ₹13.7L**
Productivity	~23 FP/person-month

USE CASE POINTS (UCP)

Step 1: Actor Weighting (UAW)

Actor	Type	Weight	Reason
Payment Gateway	System via API	Simple = 1	REST JSON calls
Fraud Analyst	GUI User	Complex = 3	Full dashboard
Simulator Engine	System via API	Simple = 1	Automated calls
Alert Recipient	External System	Average = 2	Receives notifications
Regulatory System	External via file	Average = 2	Compliance reports
ML Training System	System via API	Average = 2	Kaggle API download

$$UAW = 1 + 3 + 1 + 2 + 2 + 2 = 11$$

Step 2: Use Case Weighting (UUCW)

- Simple (≤ 3 transactions) = 5 pts
- Average (4–7 transactions) = 10 pts
- Complex (> 7 transactions) = 15 pts

#	Use Case	Complexity	Weight
UC1	Evaluate Transaction for Fraud	Complex	15
UC2	Pre-Authorization Decision (BLOCK/CHALLENGE/ALLOW)	Complex	15
UC3	Generate Explainable AI Report	Complex	15
UC4	Detect Fraud Ring via Graph	Complex	15
UC5	Send Multi-Channel Alert	Average	10
UC6	Manage Case (Analyst Workflow)	Average	10
UC7	Train ML Model	Complex	15
UC8	Run Transaction Simulation	Average	10
UC9	View Analytics Dashboard	Average	10
UC10	Configure Risk Thresholds	Simple	5
UC11	Device Fingerprint + Geolocation	Average	10

UC12	Phishing Detection	Average	10
UC13	Merchant Reputation Scoring	Simple	5
UC14	View Performance Metrics	Simple	5

UUCW = 150

UUCP = UAW + UUCW = 11 + 150 = 161

Step 3: Technical Complexity Factor (TCF)

#	Factor	Weight	Rating	Score
T1	Distributed System	2	1	2
T2	Response/Throughput	1	5	5
T3	End-User Efficiency	1	3	3
T4	Complex Internal Processing	1	5	5
T5	Code Reusability	1	3	3
T6	Easy to Install	0.5	2	1
T7	Easy to Use	0.5	3	1.5
T8	Portable	2	2	4
T9	Easy to Change	1	3	3
T10	Concurrent	1	4	4
T11	Special Security	1	5	5
T12	Direct 3rd Party Access	1	3	3
T13	Special User Training	1	2	2
Sum				**41.5**

TCF = 0.6 + (0.01 × 41.5) = 1.015 ≈ 1.02

Step 4: Environmental Factor (ECF)

#	Factor	Weight	Rating	Score
E1	Familiar with dev process	1.5	3	4.5
E2	Application experience	0.5	3	1.5
E3	OO experience	1	4	4
E4	Lead analyst capability	0.5	3	1.5
E5	Motivation	1	4	4
E6	Stable requirements	2	3	6
E7	Part-time workers	−1	2	−2
E8	Difficult programming language	−1	1	−1
Sum				**18.5**

$$ECF = 1.4 + (-0.03 \times 18.5) = 1.4 - 0.555 = 0.845 \approx 0.85$$

Step 5: Final Use Case Points

$$UCP = UUCP \times TCF \times ECF = 161 \times 1.02 \times 0.85 = 139.6 \approx 140$$

Step 6: Effort from UCP

Industry standard: 20 person-hours/UCP

Metric	Value
Final UCP	140
Effort @ 20 hrs/UCP	**2,800 person-hours**
Team Size	2 developers